

Horipriya Das Arpita

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 Dhaka, Bangladesh

 Horipriya-Arpita

 Horipriya

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 Portfolio

Skills

Languages: C, C++, Java, JavaScript, Python, SQL
Concepts: OOP, Design Patterns, SOLID Principles, DSA
Backend: Python, FastAPI, Django, Node.js, Express.js, REST APIs, PostgreSQL, MySQL, Redis, Docker
Frontend: React.js, Next.js, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Figma (Auto Layout, components, design systems)
AI/ML: LLM, RAG, NLP, Generative AI, Agentic AI, LoRA Fine-tuning, LangChain.js, Hugging Face, Neo4j
Integrations & Tools: Stripe, OAuth, Git, GitHub, Postman, Vercel, Crawl4AI, Playwright, Selenium

Professional Experience

Associate AI R&D

Sigmoix AI

12/2025 – Present | Remote

- Built a production ready Text-to-SQL system with AI-driven schema enrichment, a PostgreSQL/Redis backend, API key authentication, and rate limiting, shipped with an embeddable JS widget for frontend integration into any web platform.
- Developed ConCommerce, a deployed conversational AI product combining an LLM-based recommendation engine, intent detection, and backend logic for smart follow-up handling across ecommerce use cases.
- Designed agentic RAG pipelines (LightRAG, RAPTOR) powering voice agent and Text-to-SQL retrieval, optimizing backend retrieval accuracy and response latency.
- Engineered backend data collection infrastructure with bot detection bypass (Playwright stealth plugin, fingerprint overrides, human-like behavior simulation) to support reliable large scale web scraping.

AI R&D Engineer, Intern

Sigmoix AI

09/2025 – 11/2025 | Remote

- Built scalable backend data-extraction pipelines for 10K+ ecommerce products using Crawl4AI, Firecrawl, and Selenium, with multi-threaded scraping, user agent rotation, and automated preprocessing.
- Fine-tuned Gemma LLM using LoRA on 10K+ product datasets, improving domain-specific query understanding and retrieval accuracy.
- Built and tested backend vector database and Neo4j graph database query pipelines, and developed Streamlit and Gradio interfaces for evaluation and benchmarking, bridging backend retrieval systems with usable frontend tools.

Software Engineer, Intern

Luminous Lab

12/2024 – 05/2025 | Dhaka, Bangladesh

- Built and shipped Fullstack Shopify apps using Remix, Polaris, and Prisma, owning user registration and referral workflows, with REST API and Stripe payment integrations, plus webhooks and webSockets for real-time backend sync.
- Designed UI flows in Figma for Carlo and translated them into responsive React components, maintaining consistency with the design system.
- Integrated frontend dashboards with backend APIs for the Beige and Carlo platforms, ensuring smooth data flow and consistent UI behavior across the stack.
- Migrated legacy Flowise AI workflows to LangChain.js, implementing custom LLM chains and RAG with ChromaDB for contextual PDF-based Q&A, and engineered Bangla conversational prompts to optimize VoiceERP's calling based AI interactions.

Projects

Face Recognition System

AI-Based Facial Identity Matching

07/2025 – Present

- Built a full-stack web app with user registration and real-time identity verification using facial features.
- Implemented backend for known face storage, face embeddings, and recognition with automated CSV attendance logging.
- Designed responsive frontend for image uploads, live webcam detection, and instant feedback.
- Enabled duplicate prevention and timestamped daily attendance tracking for efficiency.
- Technology : Python, OpenCV, Next.js, Express.js, MySQL

AI Storybook Generator

Chapter-wise Story & Image Creator

02/2025 – Present

- Developed a complete full-stack platform where users create personalized storybooks with chapter-wise text and illustrations.
- Built API routes for story creation, image retrieval, and storage using Prisma ORM, ensuring efficient database access patterns.
- Integrated AI story + cover image generation using Gemini and Hugging Face models and Cloudinary for storage.
- Technology: Next.js, React, Prisma, MySQL, ClerkAuth

Open Source Projects

Shahad's Agent – AI Browser Extension

Open Source Collaboration (mentored by a Senior Software Engineer)

09/2025 – Present | Remote

- Designed and refactored a SubAgent-based agentic architecture, enabling autonomous task execution with improved observability, debuggability, and secure tool access.
- Strengthened the core AI execution and tooling framework, improving system maintainability, reliability, and test stability while preserving full execution traceability.
- Delivered end-user AI capabilities including context aware webpage chat and Gmail thread summarization, ensuring smooth, secure performance across Chrome and Firefox

Organisations

Software Engineering Society, SUST

Executive Member

02/2024 – 06/2025 | Sylhet, Bangladesh

Certificates

- ICPC Orbitax SUST SWE Technovent
- National Girls' Programming Contest 2022
- Ada lovelace Programming Competition 2022
- BdApps National Hackathon 2022

Education

B.Sc. In Software Engineering

Shahjalal University of Science and Technology

02/2020 – 07/2025 | Sylhet, Bangladesh

CGPA: 3.79